

Ventricular Assist Devices

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Tracking Information

Publication Number

100-3

Manual Section Number

20.9.1

Manual Section Title

Ventricular Assist Devices

Version Number

2

Effective Date of this Version

12/01/2020

Implementation Date

07/27/2021

Description Information

Benefit Category

Inpatient Hospital Services

Prosthetic Devices

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

A ventricular assist device (VAD) is surgically attached to one or both intact ventricles and is used to assist or augment the ability of a damaged or weakened native heart to pump blood. Improvement in the performance of the native heart may allow the device to be removed.

Indications and Limitations of Coverage

B. Nationally Covered Indications

1. Post-cardiotomy (effective for services performed on or after October 18, 1993)

Post-cardiotomy is the period following open-heart surgery. VADs used for support of blood circulation post-cardiotomy are covered only if they have received approval from the Food and Drug Administration (FDA) for that purpose, and the VADs are used according to the FDA-approved labeling instructions.

2. Left ventricular assist devices (LVADs) are covered if they are FDA approved for short-term (e.g., bridge-to-recovery and bridge-to-transplant) or long-term (e.g., destination therapy) mechanical circulatory support for heart failure patients who meet the following criteria:

- Have New York Heart Association (NYHA) Class IV heart failure; and
- Have a left ventricular ejection fraction (LVEF) $\leq 25\%$; and
- Are inotrope dependent

OR

have a Cardiac Index (CI) < 2.2 L/min/m², while not on inotropes, and also meet one of the following:

- Are on optimal medical management (OMM), based on current heart failure practice guidelines for at least 45 out of the last 60 days and are failing to respond; or
- Have advanced heart failure for at least 14 days and are dependent on an intra-aortic balloon pump (IABP) or similar temporary mechanical circulatory support for at least 7 days.

Beneficiaries receiving a VAD must be managed by an explicitly identified, cohesive, multidisciplinary team of medical professionals with appropriate qualifications, training, and experience. The team embodies collaboration and dedication across medical specialties to offer optimal patient-centered care. Collectively, the team must ensure that patients and caregivers have the knowledge and support necessary to participate in informed decision making. The team members must be based at the facility and must include individuals with experience working with patients before and after placement of a VAD.

The team must include, at a minimum:

- At least one physician with cardiothoracic surgery privileges and individual experience implanting at least 10 durable, intracorporeal, left ventricular assist devices over the course of the previous 36 months with activity in the last year.
- At least one cardiologist trained in advanced heart failure with clinical competence in medical- and device-based management including VADs, and clinical competence in the management of patients before and after placement of a VAD.
- A VAD program coordinator.
- A social worker.
- A palliative care specialist.

Facilities must be credentialed by an organization approved by CMS. The process for organizations to apply for CMS approval to be designated as a credentialing organization for LVAD facilities is posted on our web site along with a list of approved credentialing organizations, approved standard versions, and credentialed facilities:

<http://www.cms.gov/Medicare/Medicare-General-Information/MedicareApprovedFacilities/VAD-Destination-Therapy-Facilities.html> 

C. Nationally Non-Covered Indications

All other indications for the use of VADs not otherwise listed remain non-covered, except in the context of Category B investigational device exemption clinical trials (42 CFR 405) or as a routine cost in clinical trials defined under section 310.1 of the National Coverage Determinations (NCD) Manual.

D. Other

This policy does not address coverage of VADs for right ventricular support, biventricular support, use in beneficiaries under the age of 18, use in beneficiaries with complex congenital heart disease, or use in beneficiaries with acute heart failure without a history of chronic heart failure. Coverage under section 1862(a)(1)(A) of the Act for VADs in these situations will be made by local Medicare Administrative Contractors within their respective jurisdictions.

Transmittal Information

Transmittal Number

10737

Coverage Transmittal Link

<https://www.cms.gov/files/document/r10837NCD.pdf> 

Revision History

07/2025 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to NCDs. No policy is being changed as a result of these updates. ([TN 13327](#) ) (CR14041)

06/2025 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to NCDs. No policy is being changed as a result of these updates. ([TN 13251](#) ) (CR14041)

01/2024 - Transmittal 12318 issued October 19, 2023, is being rescinded and replaced by Transmittal 12441, dated January 3, 2024, to add FISS to BR 13390.11 and to add clarifying verbiage to business requirement 13390.12. All other information remains the same. ([TN 12441](#) ) (CR13390)

10/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12318](#) ) (CR13390)

08/2022 - Transmittal 11545 dated August 5, 2022, is being rescinded and replaced by Transmittal 11584, dated, August 31, 2022. ([TN 11584](#) ) (CR12822)

08/2022 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates to specific NCDs. ([TN 11545](#) ) (CR12822)

01/2022 - Transmittal 11068, dated October 21, 2021, is being rescinded and replaced by Transmittal 11179, dated, January 12, 2022 to revise the attachment for NCD 110.24, CAR-

T, to add business requirement 12480.10.1 by adding generic unspecified procedure codes, to clarify coverage and claims processing in the policy section and to review the implementation date. All other information remains the same. ([TN 11179](#)) (CR12480)

10/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11068](#)) (CR12480)

06/2021 - The purpose of this Change Request is to inform contractors that effective December 1, 2020, CMS covers Ventricular Assist Devices (VADs) under certain conditions and criteria. ([TN 10837](#)) (CR12290)

02/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 2427](#)) (CR11491)

01/2018 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 2005](#)) (CR10318)

11/2017 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 1975](#)) (CR10318)

12/2015 - This change request (CR) is the 3rd maintenance update of ICD-10 conversions/updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818,

CR8109, CR8197, CR8691, & CR9087. Some are the result of revisions required to other NCD-related CRs released separately that included ICD-10 coding. Implementation date: 01/04/2016 Effective date: 10/1/2015. ([TN 1580](#)) (CR9252)

08/2015 - This change request (CR) is the 3rd maintenance update of ICD-10 conversions/updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, & CR9087. Some are the result of revisions required to other NCD-related CRs released separately that included ICD-10 coding. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 01/04/2016 Effective date: 10/1/2015. ([TN 1537](#)) (CR 9252)

08/2014 - This Change Request (CR) is effective for claims with dates of service on and after October 30, 2013; contractors shall pay claims for Ventricular Assisted Devices as destination therapy using the criteria in Pub. 100-03, chapter 1, section 20.9.1, and Pub. 100-04, Chapter 32, sec. 320. Effective date: 10/30/2013. Implementation date: 09/30/2014. ([TN 172](#)) (CR 8803)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [First reconsideration for Ventricular Assist Devices for Bridge-to-Transplant and Destination Therapy \(CAG-00432R\)](#)
- [Original Consideration for Artificial Hearts and related devices, including Ventricular Assist Devices for Bridge-to-Transplant and Destination Therapy \(CAG-00453N\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Ventricular Assist Devices	2	12/01/2020 - N/A	You are here
Ventricular Assist Devices	1	10/30/2013 - 12/01/2020	View

